B.Sc. (Honours) DEGREE EXAMINATION, MARCH 2026

FOURTH SEMESTER

BSc (Honours) ZOOLOGY

DSE:MG4DSEZGYA02 ARTIFICAL INTELLIGENCE IN BIOLOGICAL SCIENCES

Duration: 2 HOURS **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. **Google Colab is best described as**

A. A software that must be installed on a high-end computer
B. A free online notebook platform that works through a web browser
C. A paid programming language for data analysis
D. A tool used only by computer scientists

2.**A dataset in biological sciences is best defined as:**

A. A single biological measurement
B. A collection of unrelated facts
C. A systematically arranged collection of related biological information
D. A computer program used for analysis

3.Which of the following is an example of *categorical data* in a biological CSV file?

A. Body weight of an animal
B. Population size
C. Habitat type
D. Enzyme activity

4.Which AI-based data visualization tool is especially useful for teaching AI concepts in biology using visual blocks instead of coding?

A. Tableau
B. Microsoft Power BI
C. Orange
D. Flourish

5.Bayesian optimization is mainly used to:
A. Increase sample size
B. Select animals randomly
C. Suggest improved experimental conditions
D. Replace laboratory experiments

6.One ethical advantage of using AI in experiments is that it:
A. Increases animal usage
B. Eliminates controls
C. Reduces unnecessary use of animals
D. Avoids data analysis

7.Which of the following is an example of biological big data?
A. A single laboratory notebook
B. DNA sequences from entire populations
C. One animal observation
D. A hand-drawn food web

8.The Human Genome Project was completed in:
A. 1980
B. 1990
C. 2003
D. 2015

9.Which tool is mainly used for visualizing biological interaction networks using nodes and edges?
A. BLAST
B. Cytoscape
C. QuPath
D. Excel

10.Systematic errors in experiments mainly arise due to:
A. Natural biological variation
B. Random environmental fluctuations
C. Miscalibrated instruments or batch effects
D. Large sample size

11.Which of the following is an example of a processing error?
A. Temperature fluctuation during incubation
B. Pipetting variation
C. Incorrect background subtraction in image analysis
D. Small sample size

12.Biological big data is mainly characterized by:
A. Low volume and low speed
B. High volume, velocity, and variety

C. Only genetic information
D. Small, manually collected datasets

13.Which "omics" field studies proteins and their interactions?
A. Genomics
B. Metabolomics
C. Proteomics
D. Phenomics

14.Random errors in biological experiments are best reduced by:
A. Ignoring repeated measurements
B. Averaging multiple replicates
C. Changing instruments frequently
D. Avoiding controls

15.Machine learning helps reduce experimental error mainly by:
A. Replacing biological experiments
B. Detecting patterns and correcting inconsistencies in data
C. Increasing manual calculations
D. Eliminating the need for validation

16.In PCR gel quantification, ML-based denoising is mainly used to:
A. Increase DNA concentration
B. Remove noise and uneven illumination in gel images
C. Change primer concentration
D. Eliminate the need for controls

17. **Artificial Intelligence (AI) in biological research mainly helps in:**
A. Increasing manual calculations
B. Eliminating the need for experiments
C. Analyzing data, predicting outcomes, and optimizing experiments
D. Replacing ethical guidelines

18.Which of the following is NOT a role of AI in experimental design?
A. Variable selection
B. Adaptive experimental planning
C. Random guessing of results
D. Data analysis and validation

19.Why are CSV files widely used in biological and environmental sciences?

A. They require advanced programming skills
B. They are difficult to edit and share
C. They are lightweight, simple, and compatible with many tools
D. They can only be opened using AI software

20.Which of the following is an example of *categorical data* in a biological CSV file?

A. Body weight of an animal
B. Population size

C. Habitat type
D. Enzyme activity

(20 x1 =20)

Part B

Short Answer Questions

Answer any 10 questions

Each question carries 3 marks

21.Define Google Colab.
Mention any two features that make it suitable for Zoology students.

22.What is the role of bioinformatics in genomics?.

23. Give two examples of datasets commonly used in Zoology.

24.What is KNIME?

25.Explain the structure of a CSV file with reference to header row and data rows.

26. State any two types of projects that can be created using Teachable machine.

27.Differentiate between numerical data and categorical data in biological CSV files with one example each.

28.What are random errors and systematic errors in biological experiments?
Give one example of each.

29.How does machine learning help in reducing errors in biological experiments?
Mention any three ways.

30.Define Google Colab.
Mention any two features that make it suitable for Zoology students.

31.What is the role of bioinformatics in genomics?.

32.Differentiate between numerical data and categorical data in biological CSV files with one example each.

(10 x 3= 30)

Part C

Short Essay Questions

Answer any 4 questions

Each question carries 5 marks

33.Explain the role of beginner-friendly AI platforms in modern Zoology education.

34.Describe the applications of Artificial Intelligence in biological image analysis.

35.Describe the different types of errors encountered in biological experiments.

and explain how machine learning helps in minimizing these errors.

36.Examine the challenges of data privacy, ownership, and regulatory governance in the use of AI for healthcare ar
biological research.

37.Explain, step by step, how machine learning can be used to reduce errors in PCR gel quantification experiments

38.Describe the historical development of Artificial Intelligence applications in biological sciences.

39.Explain the sources, consequences, and mitigation strategies of algorathimic bias in healthcare.

[5x4 = 20]

PART C

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